

IH05

Ancient Indian Art, Science and the Indian Knowledge & Value System

2-3 questions a paper, and the split has been shifting towards the knowledge-system half. Stupa architecture came in 2023 and 2024. Expect match-the-following on darshana-to-founder, sanskara-to-meaning and temple-style-to-feature - all pure list recall.

Two chapters bolted together. The first half is art history - stupas, caves, temple styles, sculpture schools. The second half is the Indian Knowledge and Value System - Varna, Ashrama, Purushartha, the sixteen Sanskaras, the three Rinas, the six Darshanas. That second half is newly emphasised in the RAS syllabus and most candidates skip it. Do not skip it; it is the cheapest scoring block in Indian History because the answers are fixed lists.

Ancient Art, Science and the Indian Knowledge System

Stupas

- Bharhut - Shunga, Cunningham 1873
- Sanchi - Ashoka, Shunga, Satavahana toranas
- Amaravati - Krishna river, limestone
- Parts - anda, harmika, yashti, chhatra, vedika, torana

Rock-cut caves

- Ajanta - 30 caves, all Buddhist
- Ellora - 34 caves, three religions
- Karle - largest chaitya
- Barabar - Ajivikas
- Elephanta - Trimurti

Temple styles

- Nagara - shikhara, north
- Dravida - vimana and gopuram, south
- Vesara - Deccan hybrid
- Latina, Phamsana, Valabhi

Science

- Aryabhata - Aryabhatiya 499 CE
- Varahamihira - Brihat Samhita
- Brahmagupta - zero rules, 628 CE
- Charaka and Sushruta
- Mehrauli iron pillar

Value system

- 4 Varna - Purusha Sukta
- 4 Ashrama - Brahmacharya to Sannyasa
- 4 Purushartha - Dharma to Moksha
- 16 Sanskara
- 3 Rina, 5 Mahayajna

Knowledge system

- 6 Darshana with founders
- 6 Vedanga
- Nalanda, Takshashila, Vikramshila
- Valabhi, Odantapuri

The stupa - learn the anatomy, then the three sites

A stupa is a burial mound turned into a place of worship - it holds relics of the Buddha or of a monk, and you worship it by walking around it, not by entering it. RPSC does not ask you to appreciate a stupa. It asks you to name its parts and to match a stupa to its state, its period and its excavator. So take the anatomy first; every part is a possible one-mark question.

- Medhi - the drum or base; Anda - the hemispherical dome, the core of the stupa.
- Harmika - the railed square balcony sitting on top of the dome.
- Yashti - the mast rising from the harmika; Chhatra - the tiers of stone umbrellas on it.
- Vedika - the stone railing around the stupa; Torana - the carved gateway in the railing.
- Pradakshina patha - the circumambulatory path you walk on, clockwise.

The three great stupas - site, state, period, the fact that gets asked

Stupa	State	The examinable fact
Bharhut	Madhya Pradesh (Satna)	Shunga period; chief donor King Dhanabhuti; discovered by Alexander Cunningham, 1873; sculptures now in the Indian Museum, Kolkata
Sanchi	Madhya Pradesh (Raisen)	Brick stupa by Ashoka, enlarged and stone-cased under the Shungas, four toranas under the Satavahanas; Ashokan pillar with the Schism Edict; UNESCO 1989
Amaravati	Andhra Pradesh	On the Krishna river; Satavahana period; white limestone, called Palnad marble; narrative panels
Dhamek, Sarnath	Uttar Pradesh	Gupta period; marks the site of the first sermon

ASKED IN RAS

RAS Pre 2023 and again in 2024 asked who discovered the Bharhut stupa in 1873 - answer Alexander Cunningham. The 2023 paper also asked the period and donor of its railings and gateway - Shunga period, King Dhanabhuti - and which river the Amaravati stupa stands on - the Krishna. Three separate facts, one site cluster. Learn the whole cluster.

ASKED IN RAS

RAS Pre 2023-24 set a statement question on the Great Stupa at Sanchi: the original brick stupa was built by Ashoka, it was enlarged and encased in dressed stone in the Shunga period, and the four carved toranas were added under the Satavahanas. All three statements are correct. Note the three-dynasty sequence - it is the whole answer.

TRAP

Bharhut is in Madhya Pradesh, not Bihar. The Dhamek stupa at Sarnath is in Uttar Pradesh, not Bihar - Sarnath is near Varanasi. RPSC has used exactly these two state-swaps in NOT-correctly-matched questions.

Rock-cut caves - Ajanta, Ellora and the rest

Rock-cut architecture means cutting a whole building out of a living rock face, from the top downwards, with no separate blocks assembled. Two words you need: chaitya is the prayer hall with the stupa at the far end, vihara is the monastery with cells for monks. Ajanta and Ellora are both in Maharashtra, both UNESCO sites from 1983, and both are asked almost every other year - but they are completely different in religion, which is the point RPSC exploits.

- *Ajanta - 30 caves in the Waghora river gorge, Aurangabad district, exclusively Buddhist.*
- *Two phases - phase I Satavahana and Hinayana, phase II under Vakataka Harisena, c. 460-480 CE.*
- *Rediscovered in 1819 by Captain John Smith of the British army, while hunting.*
- *Paintings - Padmapani and Vajrapani Bodhisattvas in Cave I; Mahajanaka and Chhadanta Jataka scenes.*
- *Technique is tempera on a lime-plaster ground, not true fresco - a favourite trick option.*
- *Ellora - 34 caves: Caves 1-12 Buddhist, 13-29 Hindu, 30-34 Jain. That is 12, 17 and 5.*
- *Cave 16 is the monolithic Kailasa temple of the Rashtrakuta king Krishna I.*
- *Cave 10 is the Vishvakarma chaitya, the carpenter's cave.*
- *Ellora is the only site where all three religions carved side by side - that tolerance is the exam line.*
- *Remember the split - Ajanta is only Buddhist, Ellora is Buddhist, Hindu and Jain; both are in Maharashtra and both UNESCO 1983.*
- *There are two Udayagiris - Kharavela's in Odisha with the Hathigumpha inscription, and the Gupta one near Vidisha in MP with the Varaha panel.*

The other rock-cut sites you must be able to place

Site	State	The one fact
Karle	Maharashtra	Largest chaitya hall in India; Satavahana
Bhaja and Bedsa	Maharashtra	Among the earliest chaityas
Bagh caves	Madhya Pradesh	Painted caves, close in style to Ajanta
Barabar caves	Bihar	Oldest rock-cut caves; made under Ashoka, gifted to the Ajivikas

Site	State	The one fact
Udayagiri-Khandagiri	Odisha	Jain caves of Kharavela; Hathigumpha inscription
Udayagiri near Vidisha	Madhya Pradesh	Gupta period; the great Varaha panel
Elephanta	Maharashtra	Trimurti (Maheshmurti); UNESCO 1987

Temple styles - Nagara, Dravida, Vesara

From the Gupta period onwards Indian temple building settles into regional idioms, and the textbook classification is threefold. The simplest way to hold it: the north builds a curving tower over the sanctum and leaves the compound open; the south builds a stepped pyramid over the sanctum and puts a huge gateway in the wall around it; the Deccan mixes the two. Learn the vocabulary, because the question is always feature-to-style.

Nagara against Dravida - the contrast RPSC asks

Nagara (North)	Dravida (South)
Curvilinear shikhara over the sanctum	Pyramidal vimana over the sanctum
Usually no boundary wall or gateway	Walled enclosure with towering gopurams
Tallest structure is over the garbhagriha	In later temples the gopuram outgrows the vimana
No large temple tank as a rule	Temple tank is normal
Amalaka and kalasha crown the shikhara	Stupika or octagonal crown on the vimana

- Nagara sub-types - Latina or rekha-prasada (single curved tower), Phamsana (broader, lower, straight slopes), Valabhi (barrel-roofed).
- The Odisha idiom uses its own words - deul for the tower, jagamohana for the assembly hall.

- Khajuraho temples were built by the Chandelas; the Modhera Sun temple and Dilwara by the Solankis.
- Vesara is the Deccan hybrid - associated with the Chalukyas of Kalyani and the Hoysalas.

ASKED IN RAS

RAS Pre 2023 set a match-the-following on temple styles: Nagara with curvilinear shikhara usually without a boundary wall, Dravida with a pyramidal vimana within a walled enclosure with gopurams, Vesara as the hybrid form associated with the Hoysalas. Read the feature, not the name - that is how the question is set.

Sculpture schools, Gupta art and coins

Three sculpture schools grew up in the early centuries CE, and each is identified by its stone and its style, not by its subject. Gandhara used grey schist and looked Greek. Mathura used spotted red sandstone and looked Indian. Amaravati used white limestone and told stories in panels. Gandhara and Mathura both produced the earliest images of the Buddha, roughly at the same time - that simultaneity is the statement RPSC likes to test.

The three schools - stone, style, patron

School	Material and style	Patron and region
Gandhara	Grey schist; Greco-Roman realism, wavy hair, drapery folds	Kushanas; north-west, Peshawar-Taxila
Mathura	Spotted red sandstone; indigenous idiom, robust figures	Kushanas; Mathura, Uttar Pradesh
Amaravati	White limestone; crowded narrative panels	Satavahanas; Krishna valley, Andhra

- Gupta art - Sarnath Buddha in the Dharmachakra-pravartana pose is the classic image.
 - Dashavatara temple at Deogarh - an early Nagara stone temple.
 - Bhitargaon - the finest surviving Gupta brick temple.
 - Sultanganj bronze Buddha - a large Gupta bronze.
 - Ajanta Caves 16 and 17 paintings belong to this classical phase.
-
- Numismatics is the study of coins - and coins are a standard RPSC one-liner.
 - Punch-marked karshapana coins are the earliest Indian coins.
 - Indo-Greeks were the first to issue coins bearing the king's portrait and name.
 - Kushanas issued the first large gold coins in India.
 - Gupta gold coin was the dinara; Gupta silver coinage begins only after the Shaka conquest.

Ancient Indian science - mathematics, astronomy, medicine, metallurgy

This block is pure matching - name to work to contribution. Two cautions before the table. First, Aryabhata said the earth rotates on its axis and that eclipses are caused by shadows, not by Rahu and Ketu; that is the statement question. Second, do not confuse Brahmagupta with Bhaskara II - Brahmagupta is 628 CE and gave the rules for zero and negative numbers, Bhaskara II is much later and wrote the Siddhanta Shiromani.

Scientist to work to contribution

Name	Work	Contribution
Aryabhata (b. 476 CE)	Aryabhatiya, 499 CE	Earth rotates on its axis; eclipses by shadow; pi as 3.1416; place-value
Varahamihira (6th century)	Brihat Samhita, Panchasiddhantika, Brihat Jataka	Encyclopaedia; five astronomical systems; astrology; a Navaratna of Chandragupta II
Brahmagupta	Brahmasphutasiddhanta, 628 CE; Khandakhadyaka	First systematic rules for arithmetic with zero and negative numbers
Bhaskara II	Siddhanta Shiromani	Its four parts include Lilavati and Bijaganita

- Charaka Samhita - medicine; Charaka is traditionally placed at Kanishka's court.
- Sushruta Samhita - surgery; rhinoplasty, cataract operations, the eight branches of Ayurveda.
- Vagbhata wrote the Ashtanga Hridaya.
- Nagarjuna wrote the Rasaratnakara, on alchemy and metallurgy.
- Do not pair Sushruta with the Ashtanga Hridaya - that is Vagbhata's book, and the swap is a set question.
- The Mehrauli iron pillar stands in the Qutb complex, Delhi.
- It carries a Sanskrit praise of King Chandra, identified with Chandragupta II.
- It has resisted rust for about 1600 years - the standing proof of ancient Indian metallurgy.
- Sulbasutras, part of the Kalpa Vedanga, contain the earliest Indian geometry - written to lay out Vedic fire altars.

The Indian Value System - Varna, Ashrama, Purushartha

Now the second half of the chapter. The classical Indian scheme organises a life along three axes at once - who you are in society (Varna), where you are

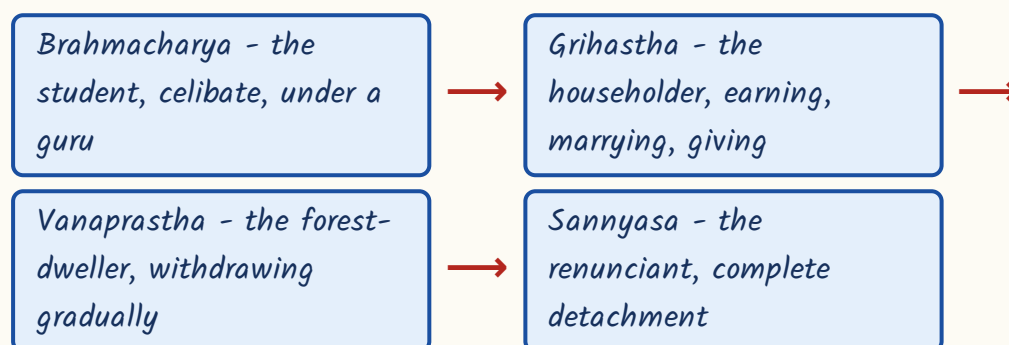
in your own lifetime (Ashrama), and what you are pursuing (Purushartha). Four in each. That symmetry is the whole design, and it is why examiners love it: the answers are closed lists that cannot change. Learn them as lists, in order, and you will never lose these marks.

- Varna - the fourfold order, first mentioned in the Purusha Sukta of the tenth mandala of the Rigveda.
- Brahmana from the mouth, Kshatriya from the arms, Vaishya from the thighs, Shudra from the feet.
- Varna is fourfold and theoretical; jati is the local birth-group and runs into thousands.
- The four Purusharthas - Dharma, Artha, Kama, Moksha - in that order.
- Dharma regulates the pursuit of Artha and Kama; Moksha is the final aim.
- Dharma, Artha and Kama together are called the trivarga.

TRAP

Varna is not jati. Varna is the four-fold classical order named in the Purusha Sukta; jati is the actual endogamous group on the ground and there are thousands of them. RPSC sets statement questions that swap these two words.

The four Ashramas - ideally 25 years each, in this order



- Grihastha is treated as the support of the other three - the student, the forest-dweller and the renunciant all live off the householder's giving.

- That is why the Grihastha ashrama is called indispensable, not merely one stage among four.
- Entry into Brahmacharya is marked by the Upanayana sanskara; its close by Samavartana.
- The ashrama scheme is a prescription in the Dharmashastras, not a description of how everyone actually lived.

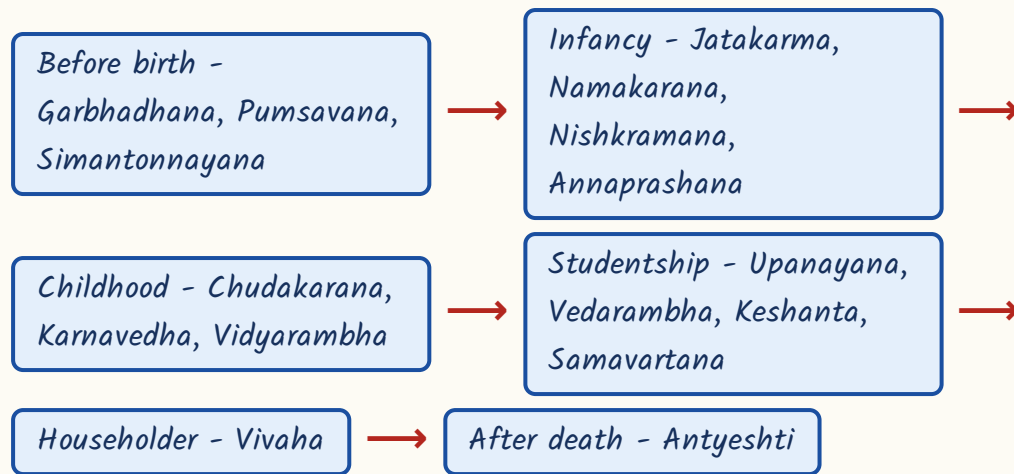
YAAD RAKHO

Two four-word chains to write on your palm. Ashrama - Brahmacharya, Grihastha, Vanaprastha, Sannyasa: B-G-V-S. Purushartha - Dharma, Artha, Kama, Moksha: D-A-K-M. Order is the question; the words alone are not enough.

The sixteen Sanskaras, the three Rinas and the five Mahayajnas

A sanskara is a sacrament - a rite that marks and refines a stage of life. The Grihya Sutras lay down sixteen of them, the shodasha sanskara, running from before birth to after death. Learn the count first - sixteen - because that alone has been asked. Then learn the order, because it is a natural life sequence and easy to reconstruct: three before birth, then infancy, then education, then marriage, then the last rites.

The sixteen sanskaras as a life-cycle - reconstruct the order from this



Sanskaras 1 to 8 - conception to childhood

No.	Sanskara	What it marks
1	Garbhadhana	Conception
2	Pumsavana	Rite in early pregnancy
3	Simantonnayana	Parting of the hair in later pregnancy
4	Jatakarma	Birth ceremony
5	Namakarana	Naming, on the 10th to 12th day
6	Nishkramana	First outing from the house
7	Annaprashana	First solid food, around six months
8	Chudakarana	Tonsure, first cutting of the hair

Sanskaras 9 to 16 - education to the last rites

No.	Sanskara	What it marks
9	Karnavedha	Piercing of the ears
10	Vidyarambha	Beginning of learning the alphabet
11	Upanayana	Investiture with the sacred thread; entry into studentship
12	Vedarambha	Beginning of Vedic study
13	Keshanta	First shaving of the beard
14	Samavartana	Graduation - return from the guru's house
15	Vivaha	Marriage

No.	Sanskara	What it marks
16	Antyeshti	Last rites - the only posthumous sanskara

- Antyeshti is the only sanskara performed after death - that single line is a ready-made one-mark question.
- Upanayana is the pivot: it opens the Brahmacharya ashrama and formal education.
- Samavartana closes studentship; Vivaha opens the Grihastha ashrama.
- Namakarana on the 10th-12th day and Annaprashana at about six months are the two dated ones.

The three Rinas from the Taittiriya Samhita - gods paid by sacrifice, sages by study, ancestors by children

Rina	Owed to	Repaid by
Deva-rina	The gods	Yajna, sacrifice
Rishi-rina (Brahma-rina)	The sages	Svadhyaya - study and teaching
Pitri-rina	The ancestors	Raising progeny, continuing the line

- Pancha Mahayajna - the householder's five daily duties.
- Brahma yajna - study of the Vedas; Deva yajna - offerings to the gods.
- Pitri yajna - to the ancestors; Bhuta yajna - to all living beings.
- Nri or Manushya yajna - hospitality to the guest.
- Note the pairing: three Rinas are lifetime debts, five Mahayajnas are daily duties. Do not merge the lists.

Shad-darshana - the six systems of Indian philosophy

Darshana means a way of seeing. Six schools accept the authority of the Vedas and are called astika; the ones that reject it are nastika. The six always come in three traditional pairs, because each pair shares a framework - Nyaya with

Vaisheshika, Samkhya with Yoga, Mimamsa with Vedanta. RPSC asks this as a straight match-the-following: school to founder, or school to core idea. Learn the table cold.

The six darshanas - founder, basic text, core idea

Darshana	Founder	Core idea
Nyaya	Gautama (Akshapada)	Logic; four pramanas - pratyaksha, anumana, upamana, shabda
Vaisheshika	Kanada	Atomism (paramanuvada) and the doctrine of categories (padarthas)
Samkhya	Kapila	Dualism of Purusha and Prakriti; twenty-five tattvas
Yoga	Patanjali	Ashtanga yoga - yama, niyama, asana, pranayama, pratyahara, dharana, dhyana, samadhi
Purva Mimamsa	Jaimini	Authority of ritual; dharma as the object of enquiry
Uttara Mimamsa (Vedanta)	Badarayana	Brahman as the ultimate reality

- Basic texts - Nyaya Sutra, Vaisheshika Sutra, Samkhya Karika of Ishvarakrishna, Yoga Sutra, Purva Mimamsa Sutra, Brahma Sutra.
- Samkhya is the oldest of the six and is atheistic in its classical form; Yoga adds Ishvara.
- Nastika schools - Charvaka (materialist), Buddhism and Jainism; the Ajivikas are also outside the six.
- The three pairs - Nyaya and Vaisheshika, Samkhya and Yoga, Mimamsa and Vedanta.
- The Gautama of the Nyaya school is not Gautama Buddha - a standing trap option.
- Keep Kapila (Samkhya) apart from Kanada (Vaisheshika); the names are one letter apart.

YAAD RAKHO

Founders in order, as one chant: Gautama, Kanada, Kapila, Patanjali, Jaimini, Badarayana. Six names, six schools, same sequence as the standard list Nyaya-Vaisheshika-Samkhya-Yoga-Mimamsa-Vedanta.

TRAP

Samkhya's basic text is the Samkhya Karika, not the Mimamsa Sutra. RPSC has set a NOT-correctly-matched question on exactly that pairing.

How knowledge was transmitted – Vedangas, gurukul and the ancient universities

Learning in ancient India moved from the gurukul, where a student lived in the teacher's house, to the mahavihara, a large residential monastery-university with thousands of students and a formal syllabus. Takshashila belongs to the earlier type – a cluster of famous teachers rather than a single institution with degrees. Nalanda belongs to the later type, and it is the one asked most often.

The ancient centres of learning

Centre	Founder or patron	Remember this
Takshashila	Gandhara region; ancient, no single founder	Panini, Chanakya, Charaka and Jivaka linked to it; a cluster of teachers, not a chartered university
Nalanda	Founded under Kumaragupta I; Palas later	Head Shilabhadra; Xuanzang and I-tsing studied there; sacked by Bakhtiyar Khalji c. 1193; UNESCO 2016
Vikramshila	Dharmapala (Pala)	Tantric studies; strong in Buddhist logic
Somapura (Paharpur)	Dharmapala (Pala)	Great monastery, now in Bangladesh

Centre	Founder or patron	Remember this
Odantapuri	Gopala (Pala)	Earliest of the Pala foundations
Valabhi	Maitrakas of Gujarat	Rival of Nalanda; Hinayana learning

The six Vedangas – the limbs of Vedic study

Vedanga	Subject	Key name
Shiksha	Phonetics, pronunciation	-
Kalpa	Ritual; includes the Sulbasutras	Geometry of the fire altar
Vyakarana	Grammar	Panini's Ashtadhyayi
Nirukta	Etymology	Yaska
Chhanda	Metre	-
Jyotisha	Astronomy	-

- Sanskrit literature of the classical age – Kalidasa wrote *Abhijnanashakuntalam*, *Meghadutam*, *Raghuvamsha* and *Kumarasambhava*.
- Vishakhadatta wrote the *Mudrarakshasa*; Shudraka the *Mrichchhakatika*; Bhasa the earliest surviving Sanskrit plays.
- Panini's *Ashtadhyayi* on grammar; Patanjali's *Mahabhashya* is its great commentary.
- Amarasimha's *Amarakosha* is the classical Sanskrit thesaurus.

CURRENT LINK

Current link: the Indian Knowledge Systems are now official policy, not just syllabus. NEP 2020 mandates their integration into curricula, the Ministry of Education runs a dedicated IKS Division, and the new Nalanda University campus at Rajgir in Bihar was inaugurated on 19 June 2024 – a deliberate revival of the ancient mahavihara on a site UNESCO listed in 2016. Position as of September 2026. This is why the RAS syllabus now names Varna, Ashrama, Purushartha and the Darshanas explicitly – expect the weight here to rise, not fall.

REVISION RECAP

1. Stupa parts - medhi, anda, harmika, yashti, chhatra, vedika, torana, pradakshina patha.
2. Bharhut - MP, Shunga, donor Dhanabhuti, found by Cunningham 1873, sculptures in Indian Museum Kolkata.
3. Sanchi - Ashoka brick, Shunga stone casing, Satavahana toranas; Schism Edict pillar; UNESCO 1989.
4. Amaravati - Krishna river, Satavahana, white limestone; Dhamek Sarnath is Gupta.
5. Ajanta - 30 caves, only Buddhist, Waghora gorge, found 1819 by John Smith; tempera, not fresco.
6. Ellora - 34 caves, 12 Buddhist + 17 Hindu + 5 Jain; Cave 16 Kailasa by Krishna I.
7. Karle - largest chaitya; Barabar - Ashoka, given to Ajivikas; Hathigumpha - Kharavela, Odisha.
8. Nagara shikhara and open compound; Dravida vimana, gopuram and enclosure; Vesara hybrid - Kalyani Chalukyas and Hoysalas.
9. Gandhara grey schist Greco-Roman; Mathura spotted red sandstone; Amaravati white limestone.
10. Coins - punch-marked earliest, Indo-Greek portraits, Kushana first big gold, Gupta dinara.
11. Aryabhatiya 499 CE - rotation of the earth, eclipse by shadow, pi 3.1416; Brahmagupta 628 CE - rules for zero.
12. Sushruta - surgery; Charaka - medicine; Vagbhata - Ashtanga Hridaya; Nagarjuna - Rasaratnakara.
13. Mehrauli iron pillar - King Chandra, rust-free for about 1600 years; Sulbasutras - geometry of fire altars.
14. Varna from the Purusha Sukta, Rigveda tenth mandala; varna is fourfold, jati runs into thousands.
15. Ashrama order B-G-V-S, 25 years each, Grihastha supports the rest; Purushartha order D-A-K-M.
16. Sixteen sanskaras from Garbhadhana to Antyeshti; Antyeshti is the only posthumous one; Upanayana opens studentship.
17. Three Rinas from the Taittiriya Samhita - Deva by yajna, Rishi by svadhyaya, Pitri by progeny.
18. Pancha Mahayajna - Brahma, Deva, Pitri, Bhuta, Nri - the householder's five daily duties.
19. Darshana founders - Gautama, Kanada, Kapila, Patanjali, Jaimini, Badarayana; nastika are Charvaka, Buddhism, Jainism.

20. Nalanda under Kumaragupta I, head Shilabhadra, sacked c. 1193; Vikramshila and Somapura by Dharmapala, Odantapuri by Gopala, Valabhi by the Maitrakas.